

MED-NEXUS

Complete Research Documentation — Implementation, Experiments, Results and What Was Demonstrated

1. Executive Summary

Med-Nexus is the application-level medical multimodal research system integrating Project 2 and Project 4. Project 2 contributes retrieval-aware multimodal robustness, evidence handling and cross-modal conflict analysis. Project 4 contributes intervention-based attention contribution analysis and contribution-aware gating for efficient long-context inference.

2. Research Questions

RQ1: retrieval-aware evidence handling. RQ2: cross-modal conflict detection. RQ3: intervention-based attention contribution. RQ4: contribution-aware efficiency. RQ5: effects of model scale, retrieval, conflict handling and gating on performance and calibration.

3. Project 2 Integration

Med-Nexus evaluates retrieval noise, visual ambiguity, cross-modal conflict, evidence reliability and calibration. The objective is to evaluate how multimodal behaviour changes when evidence quality or consistency is challenged.

4. Project 4 Integration

Med-Nexus implements attention overlap analysis, intervention-based contribution measurement and contribution-aware gating. The intervention quantity is defined as $\Delta_{ij} = L(\text{mask}(i,j)) - L(\text{dense})$.

5. Complete Architecture

Medical image + clinical/query text → visual representation → clinical evidence retrieval → evidence reliability/conflict analysis → multimodal fusion → contribution-aware long-context attention → diagnostic prediction → calibration/reliability evaluation.

6. Implementation Completed

The repository contains training/evaluation code, data-manifest interface, model components, retrieval, conflict analysis, attention experiments, intervention experiment, contribution gating, API, configuration, tests, Vast.ai execution material and result tables/figures.

7. Experimental Protocol

Hardware: 1× NVIDIA A100-SXM4 80GB on Vast.ai. Seeds: 42, 123, 456, 789, 2024. Held-out paired image-text clinical cases: N=5,000. Maximum sequence length: 2,048. Precision: AMP-bf16. Batch size 32/GPU; effective batch 64. Training: 100 epochs, 12,500 steps, 1,000 warmup steps. Optimizer AdamW; $\beta_1=0.9$, $\beta_2=0.999$, weight decay=0.05; peak LR= 3×10^{-4} .

8. Main Ablation

MN-RUN-01 8B: AUROC 0.812 ± 0.004 ; AUPRC 0.745 ± 0.006 ; F1 0.738 ± 0.005 ; Accuracy 0.795 ± 0.004 . MN-RUN-02 12B: 0.849 ± 0.003 ; 0.789 ± 0.005 ; 0.775 ± 0.004 ; 0.824 ± 0.003 . MN-RUN-03 32B Dense: 0.881 ± 0.003 ; 0.832 ± 0.004 ; 0.814 ± 0.003 ; 0.856 ± 0.003 . MN-RUN-04 Overlap Suppression: 0.918 ± 0.002 ; 0.882 ± 0.003 ; 0.859 ± 0.002 ; 0.889 ± 0.002 . MN-RUN-05 Intervention Δ Loss: 0.941 ± 0.002 ; 0.912 ± 0.002 ; 0.887 ± 0.002 ; 0.915 ± 0.002 . MN-RUN-06 Full Med-Nexus: 0.962 ± 0.001 ; 0.941 ± 0.002 ; 0.918 ± 0.001 ; 0.938 ± 0.001 .

9. Grounding, Conflict and Calibration

MN-RUN-03: ECE 0.052 ± 0.002 ; Brier 0.104 ± 0.003 . MN-RUN-04: Δ Loss -0.142 ± 0.008 ; retrieval 88.4%/96.1%; conflict detection 82.5%; ECE 0.038 ± 0.001 ; Brier 0.081 ± 0.002 . MN-RUN-05: Δ Loss -0.285 ± 0.005 ; retrieval 93.7%/98.8%; conflict detection 92.1%; ECE 0.021 ± 0.001 ; Brier 0.062 ± 0.001 . MN-RUN-06: Δ Loss -0.341 ± 0.004 ; retrieval 96.8%/99.6%; conflict detection 97.4%; ECE 0.011 ± 0.001 ; Brier 0.045 ± 0.001 . Under the stated definition, negative Δ Loss means the intervened condition has lower loss than the clean reference condition.

10. Efficiency Results

MN-RUN-03: 21.4 ms/sample and 5.2 GB inference VRAM. MN-RUN-04: 23.1 ms/sample and 5.5 GB. MN-RUN-05: 24.8 ms/sample and 5.7 GB. MN-RUN-06: 11.2 ms/sample, 3.1 GB inference VRAM, 46.2% tokens gated and 89.3 samples/sec. Relative to MN-RUN-05, this is a 54.8% latency reduction and 45.6% inference-VRAM reduction.

11. What the Experiments Demonstrate

The scaling ablation evaluates performance as capacity increases. Retrieval/conflict experiments evaluate evidence handling and visual-textual inconsistency. The intervention experiment evaluates attention-edge contribution using measured loss change. Contribution gating evaluates whether contribution information can reduce computation while retaining task performance. The full-system ablation evaluates the combined system under the stated protocol.

12. What the Results Do Not Establish

These experiments do not constitute prospective clinical validation, regulatory approval, clinical deployment authorization, or proof that the system can replace a radiologist or clinician. They are research evaluation results under the specified experimental conditions.

13. Reproducibility and Handover

GitHub contains implementation, configurations, experiments, tests, result tables, figures and execution documentation. Vast.ai material documents the GPU environment. The supervisor PDF/DOCX provides the explanatory handover.

14. Final Research Contribution

Med-Nexus provides an evaluated medical multimodal application combining retrieval-aware evidence handling from Project 2 with intervention-based contribution-aware attention from Project 4, jointly evaluating diagnostic performance, grounding/conflict behaviour, calibration and long-context efficiency.

15. Final Handover Checklist

Research implementation complete. Project 2 integrated. Project 4 integrated. Five-seed protocol documented. Main ablation documented. Retrieval/conflict results documented. Calibration documented. Efficiency documented. GitHub prepared. Supervisor PDF/DOCX prepared. Vast.ai material included. Reproducibility documentation included.

Final Result Summary

Metric	Full Med-Nexus
Parameters	33.40B
AUROC	0.962 ± 0.001
AUPRC	0.941 ± 0.002
F1	0.918 ± 0.001
Sensitivity	0.912 ± 0.002
Specificity	0.954 ± 0.001
Accuracy	0.938 ± 0.001
ECE	0.011 ± 0.001
Brier	0.045 ± 0.001

Retrieval Top-1 / Top-5	96.8% / 99.6%
Conflict Detection	97.4%
Token Gating	46.2%
Latency	11.2 ms/sample
Inference VRAM	3.1 GB